



TRIAL PLAN

Remote Sensor Monitoring of an IoT sensor network: Using low power Digital Communications

ABSTRACT

The proliferation of IoT devices sees their use spread into many harsh environments. The harsher the environment the harder the exfiltration of the collected data becomes. The Danau Girang Field Centre (DGFC) is in the Malaysian rain forest which is a very harsh RF environment as the humid ecosystem attenuates radio signals. This paper describes a set of planned experiments that will inform the design of a robust low power and low bandwidth IoT communications system to form a remote sensor network. It describes experiments that test the viability of Near Vertical Incidence Skywave (NVIS) communications in the jungle, power levels required, and explores options for the best frequency choices. A description of the planned experiments, the equipment required, system configuration, and measurements to be taken are all explored.

Mark Butterworth

Supervisors: Charith Perera, Omer Rana, Benoît Goossens, Pablo Orozco Ter Wengel

Table of Contents

Introduction	2
Aim	3
Experiment 1 Aim.....	5
Experiment 2 Aim.....	5
Experiment 3 Aim.....	5
Equipment.....	6
Antennas	6
Central Receiving Node	8
Central Receiver	9
Central Node Computer.....	9
Remote Transmitting Node.....	11
Materials	13
Central Node	13
Remote Node 1 and 2	13
Remote Node 3	14
Software.....	14
General Equipment	14
Licensing conditions	16
License Application.....	18
Health and Safety	18
RF Safety.....	18
Electrical Safety.....	18
General Safety.....	19
Log Book.....	19
Trial Methods.....	20
Experiment 1 - NVIS Signal coverage DGFC.....	20
Experiment 2 – Power level for Link Margin	22
Experiment 3 – Point to Point range testing	23
Expected Results	25
Lessons Learned.....	25
References	25

Introduction

Sensor networks are becoming more prevalent as IoT devices monitor an increasing amount of information sources. In most cases, the device data can be processed at the sensor or relayed back to base using high bandwidth, low latency techniques. The backlink could be via Wi-Fi or 4/5G cellular network communication links. For instances where these techniques are unavailable or consume too much power, other technologies, such as LoRaWAN or SigFOX, are commercially available and will backhaul data over single hops of typically 10Km in a rural environment with a line of sight between devices. Options for extending the range include using a mesh of interconnected LoRaWAN gateways. However, some situations still call for low power, long-range communications that these techniques cannot fulfil. Examples of these problems include:

- The monitoring of ubiquitous sensor networks in Antarctica(Maso et al. 2020).
- Natural environment monitoring across large areas of North Wales (Nundloll et al. 2019), again, the distances are large, and the terrain encompasses several hills making line of sight communications impossible without having relay nodes on top of each mountain.
- The long-distance transmission of IoT sensor data across the Amazonian jungle (Rey-Moreno et al. 2011). The rain forest is a harsh Radio Frequency (RF) environment consisting of dense damp vegetation, which attenuates the signal prohibiting the use of low power devices. Studies undertaken by the United States Army during the Vietnam war (Number 1971) show excessive path losses for ground-wave and line-of-sight propagation caused by dense jungle growth and tall trees.

There are several satellite services which would allow long-range IoT sensor communication. These include Argus, Iridium, and Global Star. Satellite data services are expensive to run both financially and on power utilisation. Most satellite services require a clear view of the sky to provide a good link margin, allowing a reliable signal-to-noise ratio.

The primary hypothesis of my research is that low power, low bandwidth digital signals can be used with the NVIS effect to provide a reliable long distance IoT sensor communication medium in harsh RF environments, or over ranges where traditional IoT sensor networks are not suitable. The secondary hypothesis is that a uniform signal will provide blanket coverage of the nature reserve. To prove the assumptions and claims made in year one, a trial will be held at the Danu Girang Field Centre in Malaysia. This document is the plan detailing three experiments that will be run.

The three experiments will prove that a Near Vertical Incidence Skywave signal will penetrate the jungle treetop canopy successfully and reliably to provide a resilient communications link. Trials will show that a uniform signal can be received or transmitted over the whole of the nature reserve, proving that distance from the field centre is not a defining factor in trap location. For the final system to be practical, it must operate for up to one month without a battery change. The last experiment will prove that the minimum transmitted power provides a reliable communication path at 1 Watt or less.

Aim

The trial in Malaysia aims to prove the hypothesis that a Near Vertical Incidence Skywave signal will convey trap status information reliably over an area of jungle roughly 200 kilometres in radius. The message will be bidirectional and must penetrate the jungle canopy, be reflected by the atmosphere, and be received by a central node.

The experiments will prove the effectiveness of the NVIS phenomenon and what effect the change of environment and time has on both the upper atmosphere and the moisture contained in the jungle, which will have on the propagation of the NVIS signal.

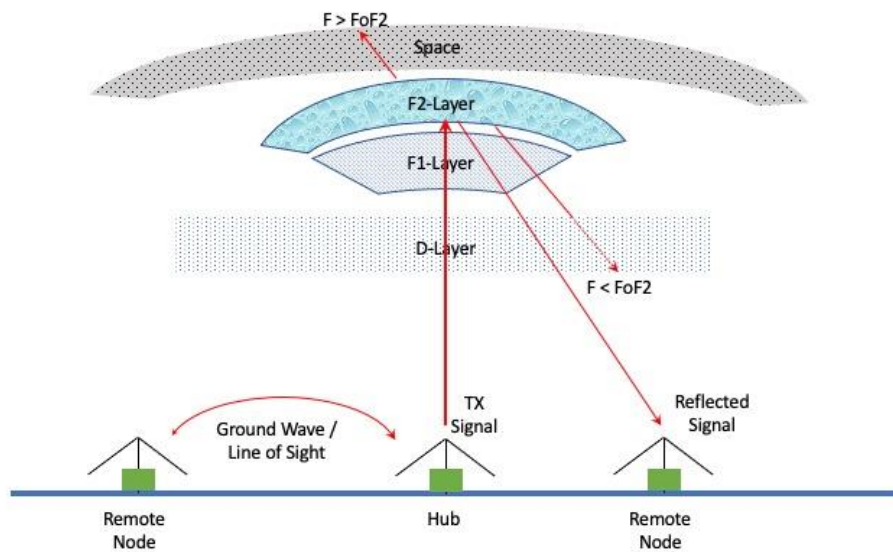


Figure 1 NVIS operation

Near Vertical Incidence skywave is a radio propagation technique. It operates by transmitting a signal almost vertically and then uses the ionosphere to reflect the signal back to earth. The coverage of the signal is like that of a spot beam from an Iridium geostationary satellite and is approximately 200 Km in diameter (Witvliet and Alsina-Pagès 2017). The ionosphere is only reflective at certain frequencies, these are in the 1MHz to 10MHz range. The best frequency is determined by atmospheric conditions as the density of the varying ionosphere layers changes as the sun ionises the gasses in the earth's atmosphere. The useable frequencies will be diurnal because sunlight excites the D-layer, increasing the attenuation of low-frequency signals. The maximum usable frequency (MUF) is the critical frequency of the F-Layer.

Using a frequency in the 3-10MHz range will facilitate a Near Vertical Incidence Skywave (NVIS) signal. The reflected signal should easily provide full coverage of the DGFC. The use of a frequency 15% below the Critical Frequency will provide reliable NVIS coverage. A global real-time ionospheric map showing f_oF2 is available from the Australian Bureau of Meteorology (Meteorology. [no date]). For use in the DGFC the angle of reflection needs to be as high as possible to ensure the signal travels as short a distance as possible and that the signal is focused on the 20-mile-wide nature

reserve. The shorter distance has the advantage of reducing the number of losses incurred when traversing the D-layer.

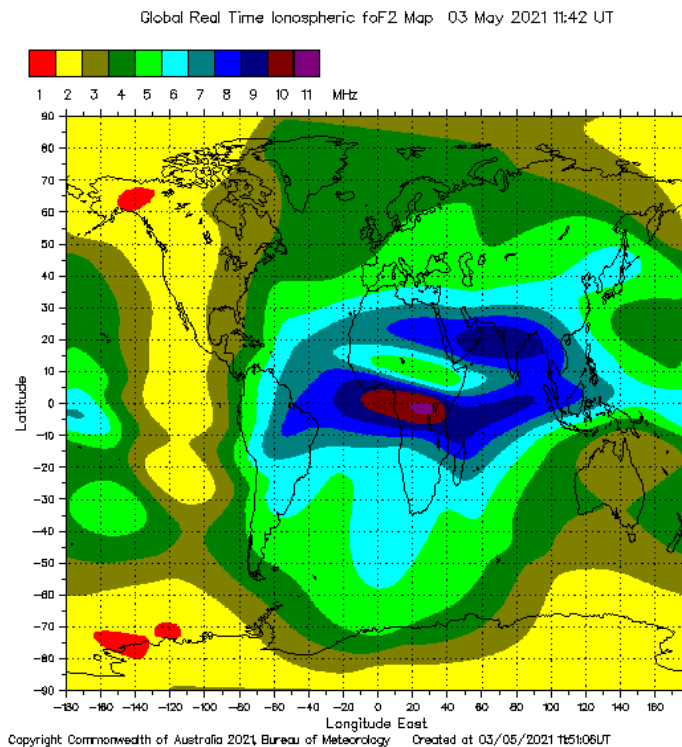


Figure 2 Global real time ionospheric foF2 map

There are three types of HF propagation, these are line-of-sight, groundwave and skywave. Line-of-sight is when the signal travels directly from the transmitter to the receiver. Line-of-sight signals are limited in distance often by obstacles or the earth's curvature. Groundwave signals travel further and use the surface conductivity of the ground to follow the earth's curvature. Groundwave propagation is dependent on frequency and the conductivity of the earth (Hughes 1992). Skywave signals are traditionally reflected off the ionosphere at low angles to facilitate long-distance communication. The angle of the reflection will constantly be changing as the atmospheric conditions alter. NVIS operates most effectively where line of sight communications is not practical.

Experiment 1 Aim

The first experiment aims to identify the optimal NVIS frequency that changes between night and day. The experiments will be run using 3 separate frequency bands 3 MHz, 5 MHz and 7 MHz. Every 15 minutes a test signal will be sent on each band. The hub computer will continuously monitor the 3 bands and record the time, signal strength and test message. The experiment will run for 24 hours to monitor variation in performance between night and day. All the units will be synchronised to GPS time and correlation with the transmission log will prove that a Skywave signal has been used rather than a ground wave.

Experiment 2 Aim

The second test will be to examine the effects of RF power levels. To preserve energy usage the system needs to operate reliably and efficiently as possible. The second test will show that the transmitted signal will provide a solid link margin. The second trial will take the results from experiment one and use the best frequencies identified, depending on the time of day, and transmit signals with the following power levels: 10 mW, 100 mW, 500 mW, 1 W, 2.5 W and 5 W. The central node computer will be receiving continuously and will record the results directly to a log file. The log file will contain the message, frequency, time, and signal to noise ratio. The test will be run over a 24-hour period to account for the changes in the atmosphere.

Experiment 3 Aim

While experiments one and two can all be done between two fixed locations, test three will need to be relocated during the experiment as it aims to prove that the NVIS signal will provide uniform blanket coverage of the DGFC reserve. With the receiving Node located centrally at the field centre, the transmitting Node will be moved further away in steps. Due to the nature of the field centre terrain these will not be a precise linear range. However, the aim will be to move away from the field centre in 1 Km steps. Also, some additional locations will be chosen that are in difficult RF locations such as behind a hill. Initially, the signal will penetrate the jungle directly and groundwave will convey the message. However, as the attenuating effects of the jungle take effect, it is expected that the skywave NVIS signal will become the dominant source. More measurements will be taken around the area where this effect occurs to ensure that the FT8 modulation is not affected by signal phasing, meaning that the signal is corrupted and lost at certain ranges. It is expected that the effects of phasing will be counteracted by the error correction contained within FT8.

Equipment

Antennas

As the signal is reflected off the ionosphere the wave is rotated shifting its phase by 90° and as the planet is rotating the received signal is shifted in frequency due to the Doppler effect. Both these naturally occurring phenomena affect the choice of antenna. The most simple but effective antenna suitable for NVIS is the inverted “V” dipole. The inverted V antenna located near the ground changes the gain and signal patterns resulting in a lobe that is focused vertically[20]. The slope of the ‘V’ affects the angle of incidence for the transmitted signal. Another enhancement to the antenna would be to add a second dipole perpendicular to the first, making a turnstile antenna. This results in a circularly polarised antenna and will compensate for the 90° phase shift in the received signal.

The angle of incidence is also important as the footprint of the reflected wave is dependent upon it. This angle can be in the range 70° - 90°, in order to reduce the skip zone. The signal for this experiment needs to be nearly vertical so an angle of 90° will be chosen (Porte et al. 2020).

During the field trials, two different antennas will be used. Both are based on an inverted “V” design, but the jungle-based antenna will be simpler to erect.

Central Antenna

The antenna will be a multiple band version that is efficient on the three NVIS bands used during this experiment.

The central node based at the DGFC will be a Tri-band Fan style dipole with 3 resonant elements operating at 3.607 MHz, 5.350 MHz and 7.070 MHz. The antenna is larger and more complex than the field antenna and will need to be supported in the centre by a telescopic aluminium pole. The longest resonant element will be a quarter-wave at 19.605 meters. Once erected the ends of the antenna will be 1.5 meters above the ground and about 42 meters apart. A centre 50Ω feed cable of RG56 coax will connect the antenna array to the receiving station.

The HF multiband dipole has 3 three dipoles that offer low impedance at their respective resonant frequency. Each element will radiate at its resonant frequency but offer a high impedance when out-of-band. The antenna will work on frequencies ±2% of the resonant frequency so will offer operating flexibility if interfering signals are present on the planned test frequency. The antenna will be constructed as described in the Stanford Research Institute’s “A field Guide to Simple HF Dipoles(Barnes et al. [1969])” (Barnes et al. [no date]).

Lowest frequency 3.6070 MHz

$$\text{Wavelength } \lambda = \frac{c}{f} = \frac{299792458}{3670000} = 81.687 \text{ m}$$

$$\text{Top Dipole Radial length} = 0.96 \frac{\lambda}{4} = \frac{81.687}{4} = 19.605 \text{ m}$$

Centre Frequency 5.350 MHz

$$\text{Wavelength } \lambda = \frac{c}{f} = \frac{299792458}{5350000} = 56.0356 \text{ m}$$

$$\text{Centre Dipole Radial length} = \frac{\lambda}{4} = \frac{56.0456}{4} = 14.009 \text{ m}$$

Highest Frequency 7.070 MHz

$$\text{Wavelength } \lambda = \frac{c}{f} = \frac{299792458}{7070000} = 42.4035 \text{ m}$$

$$\text{Bottom Dipole Radial length} = 1.01 \frac{\lambda}{4} = \frac{42.4035}{4} = 10.707 \text{ m}$$

Where:-

C = Velocity / Speed of light meters per second

f = Frequency Hertz

λ = Wavelength

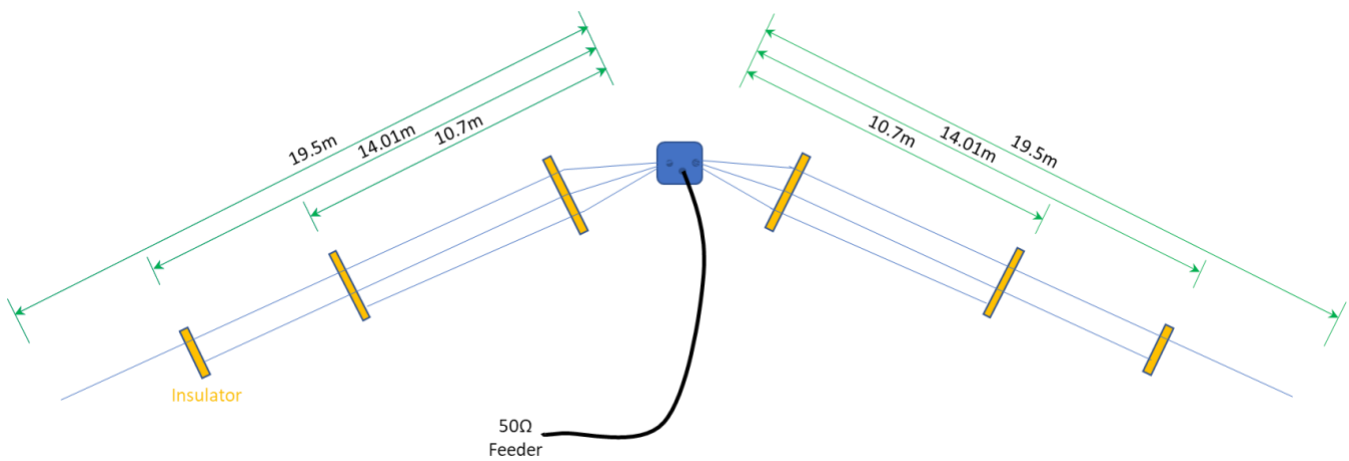


Figure 3- Fan Dipole design

Remote Node Antenna

There are three options for the transmitting / mobile node antenna. The first option is to have a separate antenna for each of the frequency bands. This would be the most efficient, but the practicalities of swapping the antennas between experiments would be physically demanding especially over a 24-hour period.

The second option would be to have a large multi-band antenna of the same construction as the central node. However, this requires a large amount of space and is cumbersome to erect.

The final option would be to have an inverted “V” dipole supported by a branch of a tree, rather than an aluminium pole. The dipole would be resonant in the central band at 5.3366 MHz making each leg of the dipole 13.365 meters long. Once erected the ends should be about 1.5 m above the ground and approximately 25 m apart. A centre 50Ω feed cable of RG56 coax at 10m will connect to the transmitter. An automatic antenna tuner will be placed in between the transmitter and antenna to provide an impedance match which will protect the transmitter from standing wave reflections and allow the most efficient transmission of signals.

The planned choice will be for an antenna designed to operate on a single frequency.

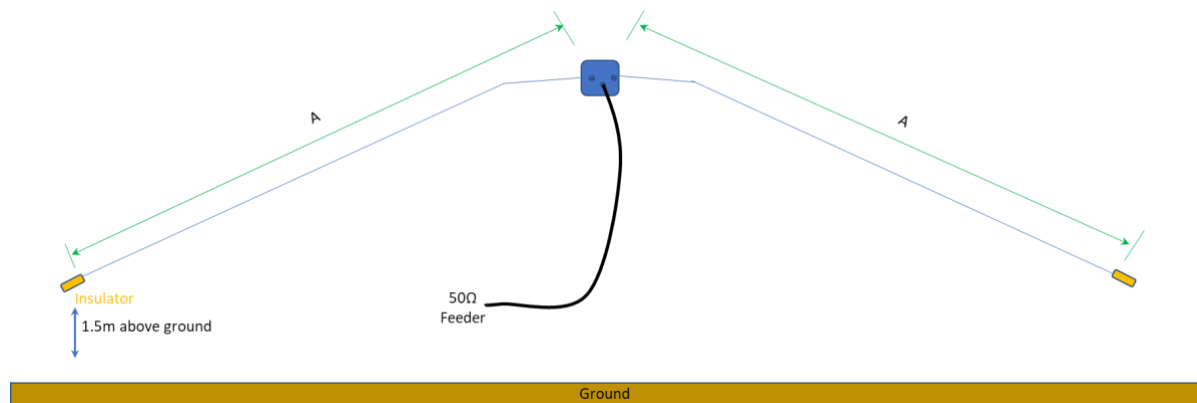


Figure 4- Inverted V Di-pole design

When an inverted V design is used it results in a lower resonant frequency. This results in the calculated radiator length being approximately 5% shorter than that of a horizontal dipole at the same frequency(Moell et al. [no date]). The lengths for the three $\frac{1}{2}$ wave antennas are shown in the table below:

Table 1- Antenna construction parameters

Frequency	Radiator Length A	Minimum central point Height	Minimum Horizontal Spread
3.6070 MHz	19.38 m	7.26 m	35.94 m
5.350 MHz	13.07 m	4.89 m	24.22 m
7.070 MHz	9.89 m	3.7 m	18.34 m

Central Receiving Node

For the experiments undertaken the receiving station will be located centrally at the Danu Girang Field Centre. The choice of a central node to record all results will aid with the automation of experiments. As the field centre has mains power the experimental setup is not restricted by its

power usage. The main receiving station will consist of a sensitive Software Defined radio, a multiband antenna, and a powerful laptop computer.

The choice of less efficient but more powerful components will enable the test system to be flexible and run continuously. All information will be automatically logged and will not rely on battery power, removing a point of failure. As the field centre has internet access all results will be automatically logged to the cloud to aid resilience. As the measurements being received are dependant upon a very accurate clock, a real time clock will be included to ensure accurate time synchronisation for gaps in GPS availability, as Internet access at the DGFC is intermittent.

Central Receiver.

The central node radio receiver will utilise an ELAD FDM-DUO.



The ELAD FDM-DUO is a direct sampling SDR transceiver that operates between 10 KHz and 54 MHz. The control software FDM-SW2 allows the operation of four simultaneous virtual receivers, each having their own dedicated audio and control channel. For these experiments three channels will be used, one on each band, 3.5 MHz, 5MHz and 7.1MHz. The audio output from each virtual channel will be fed into a separate instance of the WSJTX software, which will perform the FT8 skimming task recording all results to a CSV logfile.

Central Node Computer

The central node computer will be an intel NUC i7, with 16Gbyte memory and a 1TB hard drive. It's operating System is Windows 10 and will have Anti-Virus, FDM-SW2, WSJTX, GPS2TIME and office installed.



There will be two USB leads connecting the FDM-DUO to the computer, one contains the virtual sound card interfaces, and the other provides full computer control. To maintain accurate time synchronisation a USB GPS unit will be connected to the computer. The software package GPS2TIME will ensure accurate time if internet connectivity is lost.

Remote Desktop will be configured to enable the system to operate headless and reduce the amount of equipment required at the trial site.

Both the computer and the receiver require mains power 230V 50Hz neither components contain batteries and so power failure will result in the system shutting down and the experiment will need to be restarted.

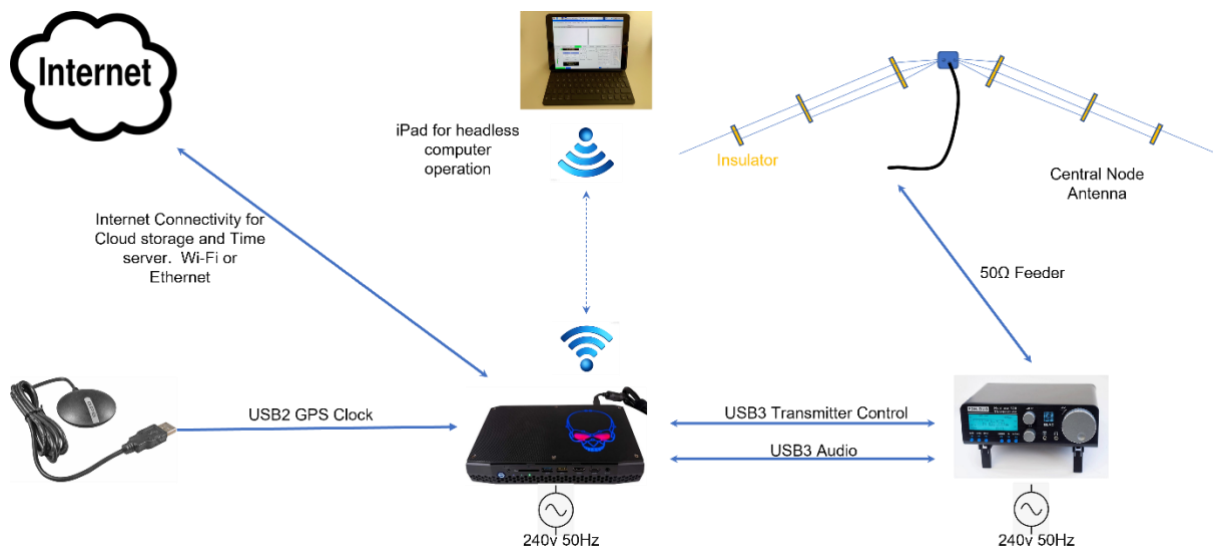


Figure 5- Central Node Diagram

Remote Transmitting Node

For the experiments remote nodes will need to be portable, easy to erect, and run for 24 hours on battery power. They will consist of an antenna, transmitter, and power source. The harsh jungle environment of high humidity, insect infestations and interest from wildlife means remote nodes will need to be well protected.

Two versions of remote node are planned:

Remote Node Version 1

The central processor will be a Raspberry Pi 4 running the “Pi OS” Operating System. The computer requires 5V at 2Amp maximum.

The transceiver will be a Hermes lite 2 Software Defined Radio also requiring 5V at 2 amps. It is a direct sampling receiver. It can output 5 Watts and is entirely software controlled. The node is configured to work autonomously once the experiment has started. The Pi OS will control the system time synchronisation. While the application WSJT-X will handle the FT8 message encoding, transmission rate, and power levels. The transceiver will be controlled using the HamRadioLib control software.

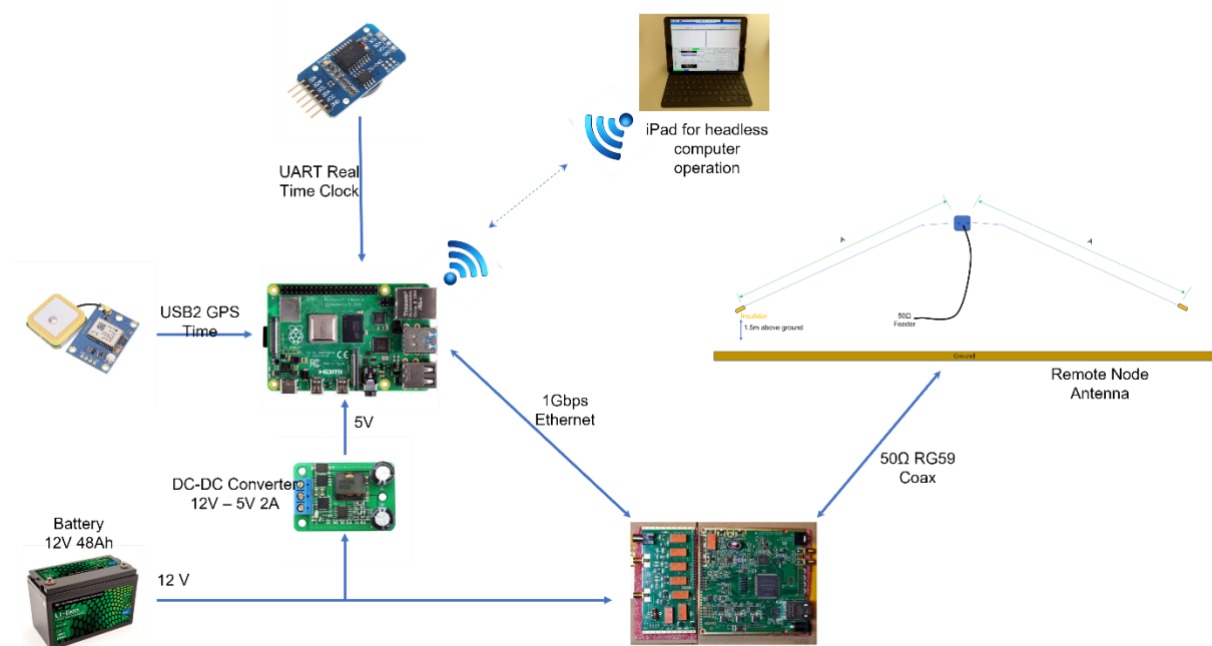


Figure 6 - Remote Node Version 1

Remote Node Version 2

The remote site has been designed to be left in the forest and be unaffected by rain or insect ingress. The remote trap module is fitted into a ruggedised 1400 Peli case. This contains the following:

- Transceiver
- Raspberry Pi with USB Audio Card, RTC, and GPS
- Battery, with 12 – 5v DC-DC converter
- Screen (for test purposes only)
- A protected BNC antenna connection

Included within the case will be Silica beads to help control the humidity within the box.

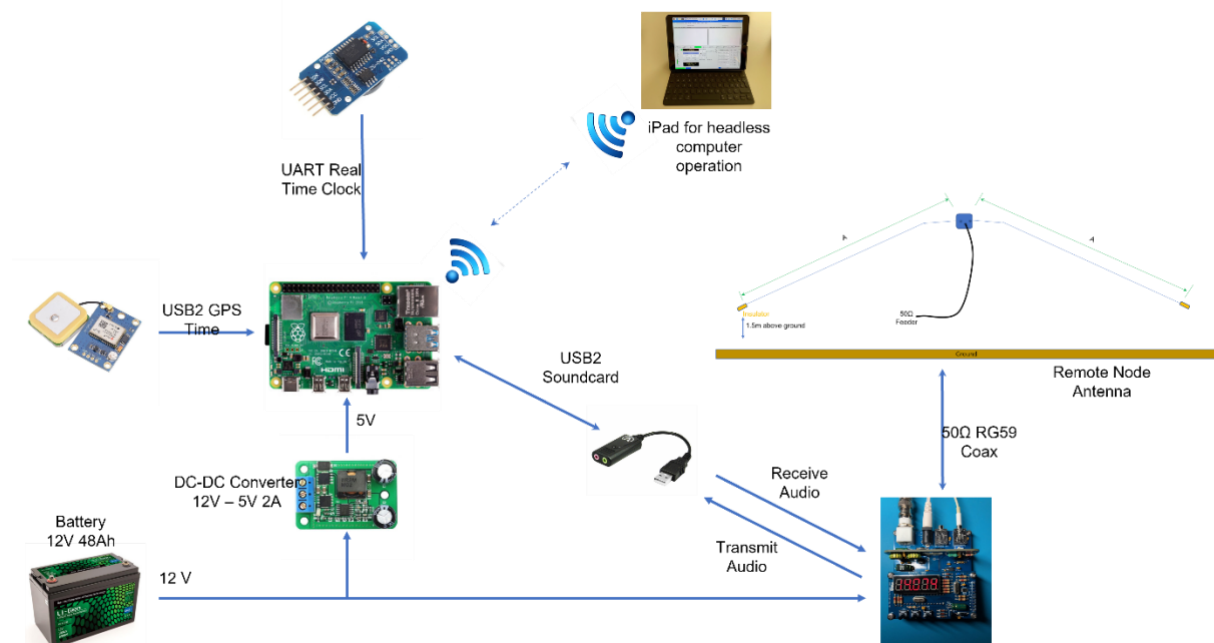


Figure 7- Remote Node Version 2

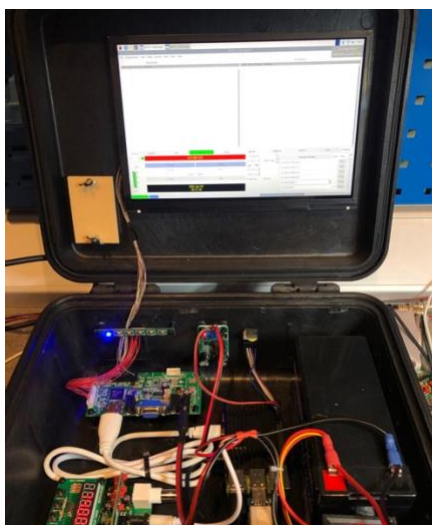


Figure 8- Remote node test setup

Materials

Central Node

Equipment	Quantity	Packed Check List
ELBIT FDM-DUO	1	
GPS receiver	1	
Intel NUC and Power Supply	1	
USB Leads	2	
USB Keyboard	1	
USB Mouse	1	
USB Hub	1	
240 Volt DC Power supply 12 V 2Ah	1	
Fan Dipole for 3.5MHz, 5MHz, and 7MHz	1	

Table 2- Central Node Equipment List

Remote Node 1 and 2

Equipment	Quantity	Packed Check List
GPS Receiver	2	
Real Time Clock	2	
Hermes Lite 2 Transceiver	2	
Raspberry Pi 4+	2	
DC-DC Converter 12V	2	
Battery 12V 24Ah Lithium	2	
USB Keyboard	1	
USB Mouse	1	
Ethernet Lead	2	
Peli Case	2	
Fixed Frequency Dipole 5 MHz	1	
Fixed Frequency Dipole 7 MHz	1	
Automatic Antenna Tuning Unit	2	
Real Time Clock	2	
SD card and Spare	4	

Table 3- Remote Node 1&2 Equipment List

Remote Node 3

Equipment	Quantity	Packed Check List
GPS Receiver	1	
Real Time Clock	1	
AFP-FSK Digital Transceiver / Red Pitaya	1	
Raspberry Pi 4+	1	
DC-DC Converter 12V	1	
Fixed Frequency Dipole 3.5MHz	1	
USB Keyboard	1	
USB Mouse	1	
USB sound Card	1	
Peli Case	1	
3.5mm Audio cables	2	
Automatic Antenna Tuning Unit	1	
SD card and Spare	2	

Table 4- Remote Node 3 Equipment List

Software

Software Package	Installed on Remote Node 1 + 2	Installed on Remote Node 3	Installed on Central Node
Pi OS	Yes	Yes	No
Windows 11	No	No	Yes
GPS2TIME	No	No	Yes
HamRadioLib	Yes	Yes	Yes
WSJTX	Yes	Yes	Yes
Office 365	No	No	Yes
VNC Server	Yes	Yes	Yes
VNC Client	No	No	No
Virtual Box	No	No	Yes
Virtual Machine Instances	No	No	Yes

Table 5- Installed Software List

General Equipment

Equipment	Quantity	Packed Check List
iPad	1	
iPad Lifeproof Case	1	
iPad charger	1	
Type G power plug adapters	2	
4 socket extension lead	1	
RF Power meter	1	
Licensing Validation Agreement	1	
A Copy of the UK licensing regulations	1	
A Copy of Malaysia's licensing regulations	1	
Nano Vector Network Analyser	1	
BNC Connectors	6	
50Ω Coax Feeder	50m	

Table 6- General Equipment List

To ensure resilience copies of each software install package and operating system will be stored on a USB stick so that any system failure can be recovered while operating in the field. Spare configured SD cards will be carried for the Raspberry Pi's and for the Red Pitaya.

The Control Node computer runs a VM (Virtual Machine) for all tests ensuring any configuration changes or system crashes that corrupt the software environment does not impact the experiment by allowing system recovery to be achieved quickly. A separate VM will be pre-created for each experiment reducing the need for in-field configuration with backup VM instances being stored on separate media.

All systems operate headless so will not require a monitor to be taken to Malaysia. All system viewing will occur using VNC remote desktop. In the field, this can be viewed on an iPad that will be housed in a waterproof LifeProof case. The case offers all-weather protection and is IP68 certified. It offers shock, dust and water ingress protection up to 2 meters for 1 hour, making it ideal for jungle operation. The case allows the remote operation of all nodes, whether in the building at the field centre or out in the nature reserve where wet humid conditions are expected while allowing all buttons to be operated. If rain showers occur, the equipment will not be affected. However, it is unlikely the touchscreen will work effectively when wet, so some form of sheltering will be required if it is raining.

Equipment Provided by DGFC	Quantity	RS Part Number	
RS PRO 12V Sealed Lead Acid Battery - 55Ah	3	727-0408	
RS PRO Battery Charger For Lead Acid 12V 14A	1	771-6276	
50Ω Coax Feeder	Dependant on DGFC		
Hand tools	Various		

Table 7- Equipment List Requested from DGFC

Licensing conditions

To transmit legally a radio transmission license must be held for the country being operated in and the frequencies on which the signals are being transmitted. Malaysia and the UK are included in the list of countries that have a reciprocal agreement. The reciprocal agreement is a common agreement that the home country's amateur license meets an agreed minimum standard. The countries with a reciprocal agreement are: Thailand, India, Switzerland, Indonesia, China, Finland, Laos, Mongolia, Germany, Vietnam, Myanmar, United Kingdom, and Czechoslovakia.

In Malaysia the Malaysian Communications and Multimedia Commission (SKMM) controls the issuing of licenses including reciprocal agreements. To operate an amateur radio station in Malaysia, the operator must have an Amateur Radio Operator's Certificate and an Amateur Station Apparatus Assignment (ASAA) or a foreign amateur radio licensed from countries that have a reciprocal arrangement with Malaysia (Malaysian Communications and Multimedia Commission 2012). (Malaysian Communications and Multimedia Commission 2012)

The International Telecommunication Union (ITU) has split the world into three regions. Each region has slightly different band plans and frequency allocations. The three regions are:

- **Region 1** - comprises Europe, Africa, the Commonwealth of Independent States, Mongolia, and the Middle East west of the Persian Gulf, including Iraq.
- **Region 2** - covers the Americas including Greenland, and some of the eastern Pacific Islands.
- **Region 3** - contains most of non-FSU Asia east of and including Iran, and most of Oceania.

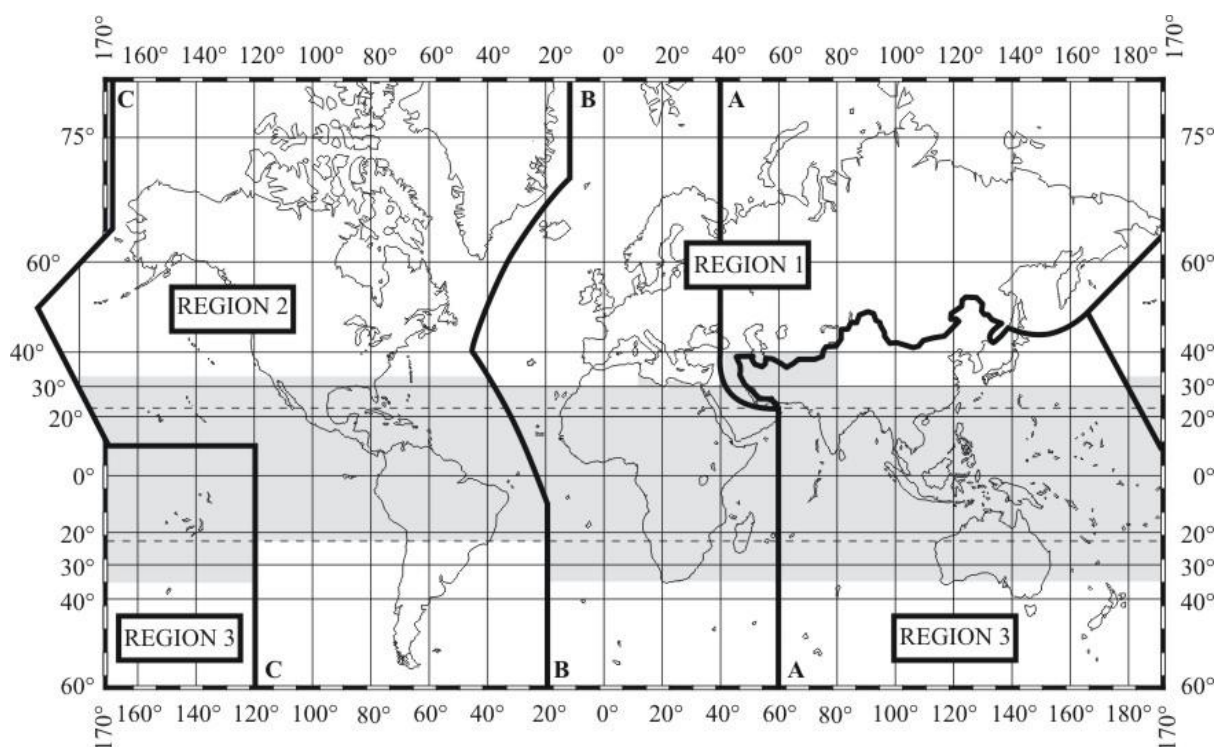


Figure 9- ITU Regions and the dividing boundaries

Malaysia is in region 3 of the ITU band plan. (Regionally harmonised bands [no date])

In Malaysia, SKMM has the overall responsibility for managing the radio frequency spectrum under the Communication and Multimedia Act 1998 (CMA98). Among other responsibilities of the SKMM include the task of developing a spectrum plan in respect of all or any part of the spectrum.

From the Malaysian Spectrum Plan 2017, the frequencies being used are shown in the table below. The table also shows the ITU sub restrictions for region 3 and the specific Malaysian restrictions. It is the Malaysian restrictions that take primacy when operating in Malaysia. (Malaysian Communications and Multimedia Commission 2017).

Frequency Band	Effective Radiated Power (ERP)	ITU Restrictions Region 3	Malaysian Restrictions
3.5000 MHz – 3.9000 MHz	400 Watts		MLA 83
5.3515 MHz – 5.3665 MHz	15 Watts	5.133B	MLA 93
7.0000 MHz – 7.1000 MHz	400 Watts	5.140 5.141 5.141A	MLA88 MLA93
7.1000 MHz – 7.2000 MHz	400 Watts	5.141A 5.141B	MLA93

Table 8- Malaysian Licensing Restrictions

The International restrictions to the ITU region 3 band plan pertinent to Malaysia are:-

5.133B Stations in the amateur service using the frequency band 5 351.5-5 366.5 kHz shall not exceed a maximum radiated power of 15 W (EIRP)

Specific Malaysian licensing restrictions required by the SKMM are:

MLA83 - The following frequencies in HF band have been identified as common Public Protection and Disaster Relief (PPDR) use in Brunei, Darussalam, Malaysia and Singapore:

3 MHz Band	6 MHz Band	11 MHz Band	14 MHz Band
3.122	6.314	11.202	14.27
3.341	6.3417	11.217	14.275
3.815	6.4501	11.23	14.293
3.925*	6.771*		14.303*
3.950*			14.325*

Table 9 - Disaster Relief Frequencies in Malaysia

* - backup frequencies

MLA88 - Technical Specification for Amateur Radio Equipment (SKMM WTS ARE). MLA88 covers the minimum technical requirements that the radio equipment must meet. These include Radio Frequency emissions including frequency drift and Effective Radiated Power levels, Electromagnetic Compatibility focusing on power supply, and electrical safety requirements, especially where 240v 13A power sources are in use.

MLA93 - Specification for Land Mobile Radio Equipment (MCMC MTSFB TC T012). Land Mobile Equipment includes base stations/repeater stations, mobile stations and handheld terminals, which

are intended for voice and/or data communication. LMR equipment shall use constant envelope angle modulation with 12.5 kHz or 25 kHz channel spacing for the analogue system.

License Application

An application for a temporary ASAA license must be made to the SKMM. The temporary license will be issued for a maximum operating duration of 1 year. Once expired another license must be applied for. The application process can take up to 3 months to process fully. Once issued the regulation states that the callsign will be 9M6/G7PRW.

Health and Safety

RF Safety

In Malaysia, amateur radio equipment shall comply with the International Commission on Non-Ionising radiation protection (ICNIRP) guidelines 2020 for limiting exposure to time-varying Electromagnetic Field (EMF) in the frequency range up to 300 GHz.

The frequencies and power levels being used for the experiment are compliant with the guidelines. This is because the average power in the band being used is $\leq 10\text{W}$ with the peak power $\leq 100\text{W}$ EIRP. The maximum power each transmitter is capable of producing is 5 Watts ensuring levels cannot be exceeded even in the event of system failure. (Ziegelberger et al. 2020)

To prevent accidental RF exposure equipment will not be powered until the antenna is fully erected, connected to the equipment under test, and personnel moved from the vicinity of the radiators.

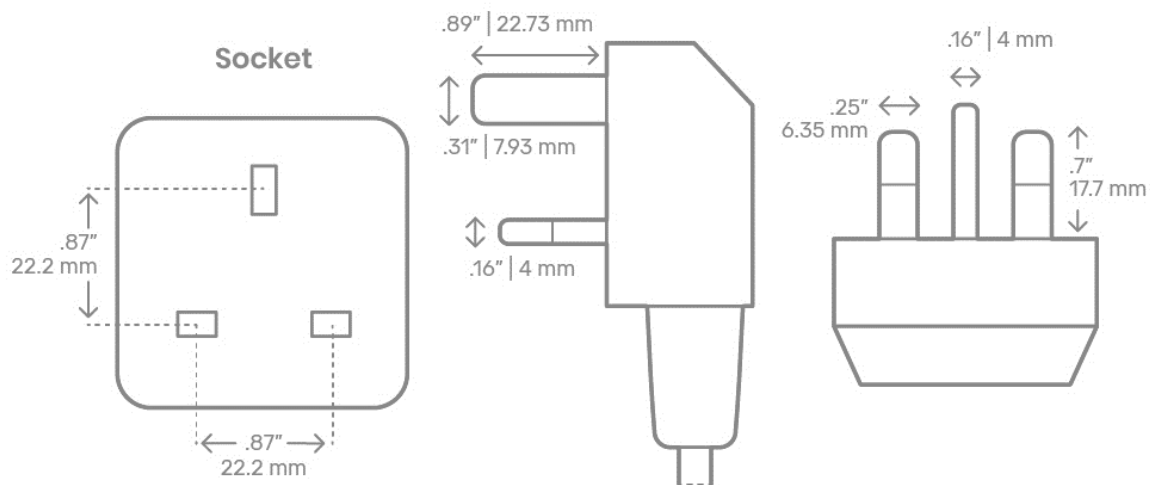
Electrical Safety

The remote nodes will be operating in harsh jungle environments which include high levels of humidity and possible water ingress to the equipment. All equipment associated with the remote nodes will operate on 12V or less and when converted to other levels will only be stepped down in voltage. Low voltage operation will remove any chance of electrical shock hazards. Equipment will be fused at the point of battery connection to ensure any accidental shorts or system failures will not result in burns or fire. Each remote node Peli case will have emergency isolation/shut off switch located on the outside of the case.

The central node will have a transceiver and a computer, both operating from a 240 V 50 Hz 13A supply. These systems will be based in the DGFC and will be inside a building. They will not be operated outside, or in wet conditions. The operating specifications for Malaysian mains voltage are 240 V $\pm 5\%$, -10% and frequency 50 Hz $\pm 1\%$ or 230 V $\pm 10\%$ and frequency 50 Hz $\pm 1\%$. The Malaysian electrical regulations (Komunikasi and Malaysia [no date]) specify power supplies must be pre-approved by the relevant regulatory body before they can be used with radio equipment. In this instance, all equipment adapters will be CE marked and commercially purchased and checks made to ensure they operate to the above specifications.

The electrical socket in Malaysia is a Type G which is the same as the UK so travel adapters or special leads will not be required.

Dimensions.Guide | Type G Plug & Socket



The Type G Plug & Socket is a 3 plug socket primarily used in UK, Ireland, Malta, Malaysia, and Singapore. The Type G Plug & Socket is grounded, has an amp rating of 13 A, voltage range of 220-240 V, and is compatible with plug type G.(Type G Plug & Socket Dimensions & Drawings | Dimensions.com [no date]).

General Safety

All jungle experiments will be governed by the DGFC risk assessment and will follow the rules and ways of working set out by the Danu Girang Field Centre. This includes jungle working, jungle travel, and insect and animal hazards.

Log Book

As a condition of the licensing agreement, a permanent record has to be made of all radio transmission activities. The logbook can be electronic and so for this experiment, an Excell spreadsheet listing all activities will be kept. The logbook must be made available for inspection by any authorised representative from SKMM. The logbook must be kept for a minimum of two years and should contain the following.(Malaysian Communications and Multimedia Commission 2012):

- Dates of transmissions
- Local time of each transmission commencement
- The frequency of each transmission
- The mode of transmission which will be FT8
- The power output
- The call sign from stations contacted. The experiment does not call for a contact to be made so this field will be included in the log in case someone tries to contact the experiment.
- Contact name. The experiment does not call for a contact to be made so this field will be included in the log, in case someone tries to contact the experiment.

- Details of test carried out.
- Location of the transmitter, so GPS coordinates of the remote Node will be recorded.
- Serial numbered records.

For the purposes of the experiment, a log entry will be made at the start and end of each experiment. Under the details of tests carried out the transmission repetition rate will be noted, i.e. 15 minutes. A separate log entry will be made for each frequency used or each time the frequency is changed. Again, a separate log entry will be made each time the power level is changed. The log will be completed on each transmitting Node and each evening consolidated into a master log which will be backed up to the cloud to protect it from corruption or loss.

All the units will be synchronised to GPS time and correlation with the transmission log will prove that a Skywave signal has been used rather than a ground wave.

Trial Methods

Experiment 1 - NVIS Signal coverage DGFC

Experiment 1 will identify the optimal NVIS frequency in Malaysia and the performance of the system when using the less-than-optimal NVIS frequencies. The NVIS effect is diurnal and so the frequency of signal reflection and its performance changes with environmental changes. The ionosphere layers change between night and day as the sun's rays excite them.

The hypothesis of this experiment is that a signal can be transmitted vertically through the jungle treetop canopy, be reflected off the ionosphere and received reliably at the central node based in the Danu Girang Field Centre.

The experiment will prove the following theories:

- A groundwave signal will be attenuated by the rainforest.
- The NVIS signal will penetrate the jungle canopy.
- The optimal NVIS frequency will change with atmospheric conditions.
- The attenuation effect of the different frequencies over time.

The NVIS effect is active over the 2 MHz – 10MHz frequency range. To identify which frequency would be best at providing the communications link, depending on the time of day, three separate frequency signals will be transmitted on the frequency bands 3.6070 MHz, 5.350 MHz and 7.070 MHz. The FT8 signal is a Gaussian frequency-shift keying (GFSK) that transmits a 50Hz bandwidth signal containing eight 6.25Hz tones. The suppressed carrier is modulated with the eight tones and is offset so that multiple signals can be received simultaneously. The offset can be between 0 Hz – 2500Hz. The offset will be set to 1KHz for this experiment on both the receiver and transmitter nodes.

The transmit power will be set to 1 Watt. The system will be configured to automatically send a test message every 15 minutes. The test message will be 9M6/G7PRW "*Maidenhead locator*" TEST.

The receiving station for the experiment will be the Intel NUC computer and ELAN FDM-DUO computer located in the DGFC. The receiving node will continuously monitor the 3 bands and record the time, signal strength and test message automatically to a log file. The log file is a text file with

the fields separated with spaces and the entries separated by new lines. Every message received or transmitted will be entered into the log file. An example of a log file entry is shown below: -

220222_192915 14.074 Rx FT8 1 0.1 1081 G7PRW: @ALLCALL TEST IO82NU

The fields of the log file are Date, Time, Frequency, Rx or TX, Operating Mode, Signal to Noise Ratio in dB, DT, Audio Frequency offset, Message.

The experiment will run for 24 hours to monitor variation in performance between night and day and which frequencies are most affected by atmospheric changes. Will the NVIS effect be enough for the system to operate successfully on one frequency even with the increased atmospheric attenuation?

All the units will be synchronised to GPS time and correlation with the transmission log will prove that a Skywave signal has been used rather than a ground wave.

There are three options for the remote nodes at the trial site and each has advantages and disadvantages: -

1. Single node and optimal antenna
2. Three complete Nodes with optimal antenna
3. Single node mid band antenna and Antenna Tuning Unit

Option 1 – Single node with optimal antenna

Choosing a single location in the nature reserve that is far enough away from the receiving node to counter the groundwave signal. The antenna for 3.670 MHz will be setup and the trial will run for 24 hours. The frequency will then be changed to 5.350 MHz, and the 5.350 MHz antenna will be erected. The trial will again be run for 24 hrs. Finally, the system will be reconfigured for the 7.070 MHz frequency and the experiment rerun for 24 hours.

The advantage of the single node approach is that each antenna location and transmitter properties are the same for each experiment. The disadvantages of the single node approach are that the atmospheric conditions will not be the same for the 36 hours. Without running the tests multiple times and taking an average reading it would be difficult to ensure consistent measurements. Running multiple sets of 36 hour tests are not practical, so assumptions would have to be made based on information from measurements made by the Australian Bureau of Meteorology (Meteorology [no date]).

Option 2 – Three nodes with optimal antennas

If three similar locations can be found with roughly a minimum of three wavelengths between them and are arranged so that they are roughly equal distance from the receiving Node, then the three transmitting nodes can operate simultaneously. Each node will operate on either 3.670 MHz, 5.350 MHz, or 7.070 MHz. Each system will have a fully tuned inverted 'V' dipole as the antenna and be located far enough away from the receiving node to counter the groundwave signal.

The advantages of this approach are that all three frequencies will be subjected to the same atmospheric approaches. It will mean that the experiment can be run in a single 24 hour period and would possibly allow time for multiple experiments to be run. The disadvantage of this approach is

that multiple sets of equipment would be required. However, given that extra equipment will need to be taken to Malaysia for resilience, this is not a big problem and if a hardware failure did arise then the experiment could revert to option 1. Another disadvantage would be that three trial sites would need to be visited.

Option 3 - Single node with mid-band antenna

A single location in the nature reserve that is far enough away from the receiving node to counter the groundwave signal would be chosen. The system would be set up with a 5.350 MHz inverted 'V' dipole antenna but would have an automatic Antenna Tuning Unit (ATU) connected between the transmitter and antenna. The software would then be set to switch between the 3 bands automatically. Three messages would be sent every 15 minutes and at the start of each transmission, the ATU would match the impedance of the antenna to the transmitter. The experiment would run for 24 hours.

The advantages of this approach would be that the experiment could be completed in 24 hours and only one trial site would be required. There are several disadvantages, these are: the ATU would take a finite time to tune each time the frequency was changed, so the efficiency of the impedance match would not be uniform over the 15-second transmission. Even though the impedance match to the transmitter would provide an efficient transfer of power into the antenna, the radiation would not be optimal. Assessing what the degradation in performance would be on the 3.670MHz, and 7.070MHz frequencies could be achieved but estimating what impact this has on the propagation characteristics of the antenna would be difficult.

The optimum solution for experiment 1 is option 2, that is, operating 3 fully independent nodes. Option 2 offers the most efficient use of time even when including the scoping of three separate trial sites. Option 2 also ensures that each frequency is subject to identical environmental characteristics. Should any equipment fail during the trial, options 1 or 3 will be the back up solution. Equipment to run all options will be taken to site, but there will be not enough redundancy to continue with option 2 in the event of hardware failure.

Experiment 2 – Power level for Link Margin

The second experiment will examine the effects of RF power levels and their effect on the signal link margin. In service the monitoring system is required to operate for one month without a battery change. To preserve energy the system needs to operate reliably and efficiently as possible. An easy way to conserve power is to transmit the lowest level of RF power that will reliably provide connectivity.

There are two options for conducting Experiment 2 that will be guided by the constraints of the nature reserve.

Option 1 - Using the analysed results from experiment 1, experiment 2 will use the most efficient frequencies identified, depending on the time of day, and transmit signals with the following power levels; 10 mW, 100 mW, 500 mW, 1 W, 2.5 W and 5 W. Three messages at each power level will be sent every 30 minutes during the periods identified as being the most permissive for each frequency. The physical setup for this test would require the use of an ATU and a mid-band (5.350 MHz) antenna. The physical setup would be the same as experiment 1 option 3. The advantage of this experiment would be that spot time periods would be chosen for tests to be performed and the messages would be sent manually. The selective approach would greatly speed up the solution. The

disadvantage would be that the set of results would be compromised by the sub-optimal antennas and the results would not be as comprehensive as a full 24-hour trial.

Option 2 – Using the same physical configuration as option 2 in experiment 1 the three nodes could be set to run for 24 hours on each of the power levels (10 mW, 100 mW, 500 mW, 1 Watt and 5 Watts). The frequencies and messages would remain the same as experiment 1. The advantage of this approach would be that the data could also be used to provide assurance of the NVIS effect results from experiment 1. The disadvantage of performing such a test would be that it would take 6 full days to complete.

Option 3 – Using the same setup as experiment 1 option 1, the system would be configured to operate on a single frequency and the messages will be sent using the most efficient time for the specific frequency as identified by experiment 1. The following power levels: 10 mW, 100 mW, 500 mW, 1 W, 2.5 W and 5 W, would be tested manually during these periods. Three messages at each power level will be sent every 30 minutes during the periods identified as being the most optimal for each frequency. The experiment would have to be rerun on subsequent days with the correct antenna selected. The disadvantage of this approach would be the lack of consistency of environmental characteristics between days. So the attenuating effects at each frequency would not be exactly comparable.

Receiving Station - The central node computer will receive continuously and record the results directly to a log file. The log file will contain the message, frequency, time and signal to noise ratio as described in the method for experiment 1.

Experiment 3 – Point to Point range testing

The hypothesis for this experiment is that the NVIS signal provides uniform coverage of up to 200 Km and hence the entire nature reserve. Experiment 3 will prove that the NVIS signal will provide blanket coverage of the DGFC reserve. The configuration of this experiment will be as follows:

The receiving node will remain centrally located at the field centre. The transmitting node will be moved away from the field centre in steps. Due to the nature of the field centre terrain, these will not be a precise linear range but will be best effort given the constraints set by the jungle. However, the aim will be to move away from the field centre in 1Km steps. Also, some additional locations will be chosen that are in difficult RF locations such as behind a hill. Initially, the signal will penetrate the jungle directly and groundwave will convey the message. As the attenuating effects of the jungle take effect, it is expected that the skywave NVIS signal will become the dominant source. More measurements will be taken around the area where this effect occurs, to ensure that the FT8 modulation is not affected by signal phasing, meaning that the signal is corrupted and lost at certain ranges. It is expected that the effects of phasing will be counteracted by the error correction contained within the FT8 protocol.

As experiment 3 requires the constant movement of the transmitting apparatus the trial will take place during daylight hours. The frequency of the transmitting station will be chosen once the results from experiment 1 have been analysed. At each test location, accurate GPS coordinates will be recorded, along with the transmission times. The system will be configured and switched on and then time will be allowed for the GPS to gain lock. Once the remote node has a GPS Synchronised time signal, a block of 15 test messages will be sent, one every 60 seconds. The transmitter power will be set to 1 Watt, and the audio offset frequency will be configured to be 1KHz. The transmission times, frequency, and message will be recorded in a log file. At the end of the experiment, an analysis of the received and transmitted log file messages will be made.

Once the results are analysed, it will be possible to identify the range where the groundwave is sufficiently attenuated, and the skywave NVIS signal becomes stronger. The following day more granular tests will be taken at approximately 200 meter intervals.

It may not be possible to reach all test sites in one day, so the trial may span more than one 24-hour period.

A Gantt chart shows the timing plan for the trial.

Table 10 - Gantt Chart of Trial

Task	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14
Travel out														
H&S + DGFC Briefing														
Experiment 1 setup														
Experiment 1 conduct														
Results Analysis														
Experiment 2 setup														
Experiment 2 conduct														
Results Analysis														
Experiment 3 Setup														
Experiment 3 conduct														
Analyse Results														
Pack Up														
Travel Back														

Expected Results

Lessons Learned

References

- Barnes, C., Hudick, J.A., Mills, M.E., Monmouth, F. and Stalsföht, D. [no date]. *A FIELD GUIDE TO SIMPLE HF DIPOLES Prepared for: United States Army Electronics Command Sponsored by the Advanced Research Projects Agency Under ARPA Order 371*. Available at: <https://apps.dtic.mil/sti/pdfs/AD0684938.pdf> [Accessed: 2 May 2022].
- Hughes, P. 1992. Understanding NVIS. *The British Journal of Administrative Management* , p. 20.
- Komunikasi, S. and Malaysia, M. [no date]. *SKMM WTS ARE Rev. 1.01:2007 TECHNICAL SPECIFICATION FOR AMATEUR RADIO EQUIPMENT*. Available at: https://www.mcmc.gov.my/skmmgovmy/files/attachments/WTS_ARE.pdf [Accessed: 2 May 2022].
- Malaysian Communications and Multimedia Commission 2012. *Malaysian Communications and Multimedia Commission AMATEUR RADIO COMMUNICATIONS GUIDELINES FOR AMATEUR RADIO SERVICES IN MALAYSIA Second Edition 01March2012*. Available at: https://www.mcmc.gov.my/skmmgovmy/files/attachments/Guideline_Amateur_Radio_Service_2012.pdf [Accessed: 2 May 2022].
- Malaysian Communications and Multimedia Commission 2017. *Malaysian Spectrum Plan ISSUED 2017*. Available at: <https://www.skmm.gov.my/skmmgovmy/media/General/pdf/Spectrum-Plan-2017i.pdf> [Accessed: 2 May 2022].
- Maso, J.M., Porte, J., Pijoan, J.L. and Badia, D. 2020. Study of NVIS channel for USN protocol definition in antarctica. *Electronics (Switzerland)* 9(6), pp. 1–21. doi: 10.3390/electronics9061037.
- Meteorology., C. of A.B. of [no date]. *Global RealTime Ionospheric foF2 Map @ www.sws.bom.gov.au*. Available at: [https://www.sws.bom.gov.au/Images/HF Systems/Global HF/Ionospheric Map/WorldIMap0.gif](https://www.sws.bom.gov.au/Images/HF%20Systems/Global%20HF/Ionospheric%20Map/WorldIMap0.gif).
- Moell, J., Allison, B., Hare, E.F. and Lau, Z.H.J. [no date]. *The ARRL Handbook 2021*. 98 Edition. ARRL.
- Number, A.D. 1971. Tactical Jungle Communications Study. (May 1960). Available at: <https://apps.dtic.mil/dtic/tr/fulltext/u2/445943.pdf>.
- Nundloll, V. et al. 2019. The design and deployment of an end-to-end IoT infrastructure for the natural environment. *Future Internet* 11(6), pp. 1–27. doi: 10.3390/fi11060129.
- Porte, J., Briones, A., Maso, J.M., Pares, C., Zaballós, A. and Pijoan, J.L. 2020. Heterogeneous wireless IoT architecture for natural disaster monitorization. *Eurasip Journal on Wireless Communications and Networking* 2020(1). doi: 10.1186/s13638-020-01793-3.

Regionally harmonized bands [no date]. Available at: <https://www.itu.int/en/ITU-R/information/Pages/emergency-bands.aspx> [Accessed: 2 May 2022].

Rey-Moreno, C., Bebea-Gonzalez, I., Foche-Perez, I., Quispe-Tacas, R., Liñán-Benitez, L. and Simo-Reigadas, J. 2011. A telemedicine WiFi network optimized for long distances in the Amazonian jungle of Peru. *ACM International Conference Proceeding Series*, pp. 1–6. doi: 10.1145/2414393.2414402.

Type G Plug & Socket Dimensions & Drawings / *Dimensions.com* [no date]. Available at: <https://www.dimensions.com/element/type-g-plug-socket> [Accessed: 8 May 2022].

Witvliet, B.A. and Alsina-Pagès, R.M. 2017. Radio communication via Near Vertical Incidence Skywave propagation: an overview. *Telecommunication Systems* 66(2), pp. 295–309. doi: 10.1007/s11235-017-0287-2.

Witvliet, B.A., Laanstra, G.J., van Maanen, E., Alsina-Pagès, R.M., Bentum, M.J., Slump, C.H. and Schiphorst, R. 2016. A transportable hybrid antenna-transmitter system for the generation of elliptically polarized waves for NVIS propagation research. *2016 10th European Conference on Antennas and Propagation, EuCAP 2016* (April). doi: 10.1109/EuCAP.2016.7481549.

Ziegelberger, G. et al. 2020. *Guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz)*. doi: 10.1097/HP.0000000000001210.